Compression-Aware Explainable Bimodal Deepfake Detection for Image and Video Analysis

Durgam Poojitha
Department of Computer Science and Engineering
Amrita School of Computing , Amrita Vishwa Vidyapeetam
Coimbatore, India
cb.sc.u4cse23320@cb.students.amrita.edu

Akshay Krishna Paramanandalayam Sajikumar
Department of Computer Science and Engineering
Amrita School of Computing , Amrita Vishwa Vidyapeetam
Coimbatore, India
cb.sc.u4cse23403@cb.students.amrita.edu

Alana Mariya P C
Department of Computer Science and Engineering
Amrita School of Computing , Amrita Vishwa Vidyapeetam
Coimbatore, India
cb.sc.u4cse23205@cb.students.amrita.edu

Kalyani H Karuvelil
Department of Computer Science and Engineering
Amrita School of Computing , Amrita Vishwa Vidyapeetam
Coimbatore, India
cb.sc.u4cse23423@cb.students.amrita.edu

Sidharth S Nair
Department of Computer Science and Engineering
Amrita School of Computing , Amrita Vishwa Vidyapeetam
Coimbatore, India
cb.sc.u4cse23443@cb.students.amrita.edu

Lalithamani N
Department of Computer Science and Engineering
Amrita School of Computing , Coimbatore
Amrita Vishwa Vidyapeetham, India
n_lalitha@cb.amrita.edu

***Abstract*—Deepfake technology is becoming a growing problem to the authenticity of digital media shared across social networks, particularly content that undergoes high levels of compression and speedy distribution. Though existing deep learning-based approaches achieve high accuracy under ideal conditions, they are often characterized by low accuracy in constrained, real-world settings and are completely uninterpretable as black-box models. To address these limitations, this project proposes Compression Aware Deepfake Explainability Technology (CADET), a compression-aware and interpretable bimodal framework for detecting deepfakes. The core approach introduces a compression-based preprocessing pipeline to simulate social media compression, allowing the models to extract robust features in a better manner. We implement a distinct bimodal architecture that utilizes XceptionNet for images and a combined EfficientNet-LSTM network to process video frame sequences individually. Furthermore, Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated to visualize explainable heatmaps of the model predictions. Experimental results demonstrate that the proposed framework significantly improves both spatial and temporal detection performance across compressed scenarios while also providing meaningful visual explanations. Ultimately, this work contributes towards building practical, transparent, and trustworthy deepfake detection systems suitable for real-world deployment in active social media environments.**

***Index Terms*—Deepfake Detection, Compressed Media Analysis, Explainable AI, Grad-CAM**

I. Introduction

The recent years have seen the explosion of generative artificial intelligence, which has enabled the development of very realistic synthetic media, popularly referred to as deepfakes [1]. As much as there are numerous positive uses of these technologies, they are also dangerous. Misinformation, reputational damage, and decreased trust of digital content by the population can be disseminated with deepfakes. There also exists the enormous financial impact deepfakes are having with losses projected to reach $40 billion in the United States by 2027 [2]. Generally, humans cannot detect a real video from a Deepfake one using their naked eye [3]. This is even more problematic on social media where pictures and videos are often condensed, scaled, and transmitted very fast. This processing causes visual distortions that complicate the process of detection systems to be reliable.

Numerous available deep learning models, in particular, convolutional neural networks (CNN) and transformer-based models have demonstrated good results in detecting deepfakes and image forgeries in controlled settings [4], [5]. These models however tend to fail when evaluating highly compressed media [6]. Their susceptibility to compression artifacts is one of the primary limitations as most models are trained on high-quality data. Consequently, they become very inaccurate when handling compressed media. Recent image-forgery detection approaches have also explored spatial attention and lightweight architectures to improve the identification of localized manipulation while supporting practical deployment [7].

The other shortcoming is that most of the methods only facilitate either images or videos, but not both. Moreover, the majority of models act as black boxes, giving predictions without giving an explanation of how such decisions were made. However, in scenarios such as media verification, both the factors accuracy and explainability are equally important.

People need to understand why the particular content has been flagged as a fake one for believing the system. Techniques like Gradient weighted Class Activation Map(Grad-CAM) [8] can help in addressing this problem as they will show which parts of the image/video influence the decisions made by the model. However, this does not always happen in practice as these techniques do not always get implemented in practice.

To solve all these problems, we propose Compression Aware Deepfake Explainability Technology (CADET), a compression aware and interpretable deepfake detector system for images and videos. The main idea is to make sure that the effect of image and video compression observed in social media is achieved during preprocessing. This ensures that the model gets more realistic features. There is a different pipeline for both images and videos. In the case of images, we will use the technique of detecting deepfake using an XceptionNet, while in the case of videos, we will use the combination of EfficientNet and Long Short Term Memory(LSTM).

In addition to the above method, Grad-CAM will be used to create visual interpretation to make the method more transparent. This will ensure that the model focuses on those parts of the image and important frames of the video which are significant.

The main contributions of this work are as follows:

- Compression-aware preprocessing pipeline which is capable of simulating social media like compression.
- A bimodal system that can handle both images and videos through the use of two distinct models.
- Visual and interpretable explanations through integration of Grad-CAM.

Overall, it is intended to narrow the gap between deepfake detection approaches and practical implementation. This is particularly in the case of social media content where compression plays a key role.

II. Motivation

The rise of generative artificial intelligence has made the production of highly convincing synthetic media more accessible than ever before. Deepfake technology, though originally intended for creative purposes, has increasingly been misused to spread misinformation, defame public figures through fabricated videos, and carry out identity fraud on social media platforms [9]. As these generative models continue to improve, the ability of humans to distinguish real content from manipulated content has become increasingly unreliable [6]. Given this growing threat, building systems that can automatically detect such manipulations has become a pressing necessity.

What makes this harder is that most detection models were never built with social media in mind. When images and videos are uploaded to social media platforms, they undergo heavy compression which introduces visual artifacts and distortions. Most detection models are trained on high quality uncompressed data and therefore struggle significantly when tested on compressed media [10], [11]. Since most models are trained on clean, high quality data, they tend to fall apart when tested on the kind of compressed media that actually circulates online.

On top of this, very few systems are built to handle both images and videos together. Most current approaches are designed to work with either spatial features from static images or temporal patterns from video sequences, but rarely both together. While some works like Nasser et al. [12] and Song et al. [13] have explored multimodal detection, a dedicated bimodal system that jointly handles image and video deepfake detection in a combined pipeline remains largely unexplored. This limits how broadly existing systems can be applied across the variety of content formats found on social media.

Perhaps the most critical gap is the absence of explainability in existing detection models. High performing CNN and transformer based architectures typically function as black box systems, producing only a real or fake label without offering any reasoning behind the decision [14]. In practical scenarios such as media verification, this is a serious limitation as analysts need to understand why content has been flagged in order to trust and act on the system's output. While studies like Sunkari and Srinagesh [15], recent work has also demonstrated the use of explainable machine learning and transformer-based approaches for detecting misleading digital content [16]. However, such techniques are inconsistently applied and rarely treated as a fundamental component of the detection pipeline. Ali et al. [17] also note that post-hoc explainability remains one of the least developed areas within the broader eXplainable Artificial Intelligence(XAI) research landscape.

To address these gaps, this paper proposes a bimodal deepfake detection system that is both compression aware and explainable. The system uses a compression based preprocessing pipeline to simulate social media like compression during training, XceptionNet for image based detection, and EfficientNet combined with LSTM for video based detection. Grad-CAM is integrated as an explainability module to generate visual heatmaps that highlight the facial regions and video frames responsible for the model's decision. The goal is to move beyond black box detection and provide a transparent, interpretable tool that is practically usable in real world social media contexts.

III. Related Works

Nasser et al. [12] focuses on detecting fake news and deepfake content on social media using multimodal deep learning. It combines textual features (using models like Bidirectional Encoder Representations from Transformers(BERT)), visual features (using CNN architectures), and in some cases audio features. The methodology involves extracting features from multiple modalities, followed by applying fusion techniques (early fusion and late fusion) and performing classification using deep neural networks. The models improve detection accuracy by capturing inconsistencies between textual and media content. It also highlights the importance of incorporating XAI to enhance transparency and build user

trust in deepfake detection systems.

Song et al. [13] addresses the problem of identity forgery and deepfake videos circulating on social media platforms. With the advancement of AI-based generative models, attackers can create highly realistic fake profile videos that can be used for impersonation, misinformation, and fraud. To solve this problem, the authors propose a deep learning framework called SFE-Net (Spatio-temporal Fusion Extraction Network) for detecting manipulated profile videos.

Haq and Malik [18] addresses the increasing threat posed by multimodal deepfakes on social media, where manipulated content combines both visual and audio elements to create highly realistic forgeries. Traditional detection methods, which often rely on a single modality, struggle to identify such sophisticated manipulations and fail to generalize across different datasets and attack types. To tackle this issue, the authors propose a novel framework called DARL, which treats deepfake detection as an audio visual synchronization problem. The core idea is that deepfakes disrupt the natural alignment between lip movements and speech, making inconsistencies in LipSync a reliable indicator of forgery.

Venkateswarulu and Srinagesh [15] proposes an explainable method of deepfake detection titled 'DeepExplain' that utilizes CNNs with LSTM to detect anomalies in a video. The LSTM allows the CNN capture and retain context from previous frames allowing it to detect anomalies in videos. DeepExplain also makes use Grad-CAM and SHAP values for providing explainability to the results obtained. It uses the DFDC dataset for training and testing. It also compared CNN, RNN, LSTM and GAN, but CNN with LSTM provided the highest accuracy.

Works like Khalid et al. [19] explore prototype based architectures where interpretability is intrinsically achieved by learning localized feature patches during training and computing spatial similarity mappings relative to those patches in the test images, thus identifying similar patches known to occur in deepfakes in the test images.

Ali et al. [17] provides an overview of the current research and trends in the field of eXplainable AI(XAI). It provides a background on XAI, common definitions and current research directions. It divides XAI techniques into four categories: · Data Explainability: To foster a better understanding of the datasets used to train models. · Model Explainability: For understanding the inner workings of models to make more optimized models in the future. · Post-hoc Explainability: Focuses on providing reasoning for explaining the decision taken by a model w/ respect to a particular input. · Assessment of Explainability: To evaluate the effectiveness of XAI techniques i.e, how well an XAI technique explains a model and how accurate it is. It also provides recommendations for software tools for XAI as well as future research directions.

IV. Methodology

A. Dataset Description

The datasets used are the 140k Real and Fake Faces dataset for images and FaceForensics++ dataset for videos. The datasets were selected on the basis of being sourced from social media and being readily available for use through the Kaggle platform.



Fig. 1. Sample fake image from image dataset.

1) Image Dataset: The image dataset consists of 140,000 total images of human faces. The dataset is divided into 70,000 real images from the image sharing platform Flickr that was collected by Nvidia. The images have been cropped to contain only the faces of individual subjects. The dataset consists of images of subjects in both indoor and outdoor conditions. The remaining 70,000 images are a subset of the 1 Million FAKE faces dataset found on Kaggle. The images were generated using StyleGAN.



Fig. 2. Sample fake video frame from video dataset.

2) Video Dataset: The video dataset consists of 6,000 total videos. The dataset is divided into 1,000 real videos sourced from YouTube. The remaining 5000 videos are fake videos generated from the 1,000 real videos using 5 different deepfake techniques namely: Deepfakes, Face2Face, FaceShifter,

FaceSwap and NeuralTextures. The dataset consist of videos with clear view of frontal features with almost no occlusion.

B. CADET Architecture

The CADET system can be seen in Figure 3. The architecture begins by preparing the dataset through resizing, normalization, and frame extraction. Following this, the extracted frames and images are passed to the Compression Based Preprocessing(CBP) module which will simulate real world compression artifacts commonly found in content uploaded to social media sites. From here, the images/frames are used to train the two models by providing the images and frame sequences to XceptionNet and EfficientNet, respectively. The video deepfake detection model makes use of an LSTM to detect abnormalities between sampled frames. The trained model is then used to generate predictions, which are then used alongside the pre-final layer of the model that made the prediction to obtain the Grad-CAM heatmap, thus providing visual reasoning for the model's prediction.

C. Compression Based Preprocessing

The next step would be to employ Compression-Based Preprocessing (CBP) to simulate real-world compression artifacts. In order to accomplish this, we first set a compression probability variable which will determine the likelihood of a given image being subjected to compression. Once an image is selected to undergo compression, more random values are selected to determine the compression quality level and the resize factor. The image is first scaled down by the resize factor and then subsequently scaled up again. Following this a `GaussianBlur` is applied to the image. The compression quality level is then used with the `imencode()` function of the `OpenCV(cv2)` image library to create a compressed JPG buffer. The compressed JPG buffer is then converted back into an image array using the `imdecode()` function of the same library. Following this, the image array is then converted to an Image object of the `PIL` library, which is then what is used during training. As such, the model will be familiar with compressed images and will be able to reliably differentiate between deepfakes and real images even when the images are compressed.

$$I_{\text{out}} = \mathcal{D}_{\text{JPEG}}\left(\tilde{I}_{\text{blur}}; q\right) \tag{1}$$

where the intermediate blurred image $\tilde{I}_{\text{blur}}$ and operations are defined as:

$$\tilde{I}_{\text{blur}} = \begin{cases} K_{5\times5} * I_{\text{res}}, & \text{with probability } p = 0.7 \\ I_{\text{res}}, & \text{otherwise} \end{cases} \tag{2}$$

$$I_{\text{res}} = \mathcal{R}_{\text{linear}}\left(\mathcal{R}_{\text{area}}(I_{\text{bgr}}, s), (W, H)\right) \tag{3}$$

with parameters:

- $s \sim \mathcal{U}(0.85, 1.0)$ represents the random downscaling scale factor.
- $\mathcal{R}_{\text{area}}$ and $\mathcal{R}_{\text{linear}}$ denote area-interpolation downscaling and bilinear upscaling back to original dimensions (W, H), respectively.
- $K_{5\times5}$ is a 5×5 Gaussian kernel with standard deviation $\sigma = 0$, and $*$ denotes the convolution operator.
- $\mathcal{D}_{\text{JPEG}}(\cdot; q)$ represents JPEG compression and decompression with quality factor $q \sim \mathcal{U}_{\text{int}}(10, 25)$ incorporating chroma subsampling.

D. Frame Extraction

A function `extract_videos_to_disk()` is responsible for extracting the frames from the mp4 videos in the video dataset and saving them to disk. The function extracts 16 video frames from a video, ensuring that they are evenly spaced.

E. Dataset Preparation

For the image dataset, the images are already split into training, testing and validation sets in the Kaggle platform. We have used the same split for our purposes i.e 70:15:15 for train:test:validation. The data is equally divided among the two classes, making the dataset balanced.

For the video dataset, all 6000 videos are taken for processing. From here, frame extraction occurs and the frames are saved into real and fake folders respectively. Subsequently, the frames are divided into train, test and validation sets in the ratio 70:15:15.

F. Model Learning

Following this, we must perform the learning phase for training the existing models (XceptionNet and EfficientNet) on the new dataset. In order to do this, we must add new layers on top of it to perform binary classification on the content as deepfake or not deepfake. Additionally, for the video deepfake detection process, we need to add an LSTM who will take the outputs given by the EfficientNet model and uses it for processing, which will then go to the classifier head. The hyper-parameters chosen for the model were based on the values from [20], with a learning rate of 1e-4 and batch size of 8 being used for the video model being the only exception due to better performance and resource constraints respectively.

G. Detection - Image

The image model chosen for detection is the Xception network [21] due to its state-of-the-art performance on the ImageNet dataset while also intended at being more computationally efficient with parameters than other well-known CNN models like VGGNet and Resnet [22]. The input tensor X_{img} is given to the pretrained Xception backbone which performs feature extraction on the input tensor to produce a 2048-dimensional feature vector z_{img}.

$$z_{img} = \text{Xception}_{\text{backbone}}(X_{img}) \in \mathbb{R}^{B_{img}\times2048} \tag{4}$$

where B_{img} is the batch size for the image model

z_{img} is given to a custom classification head to produce the raw logits. The classification head for the image model consists of a fully connected layer that reduces the number of

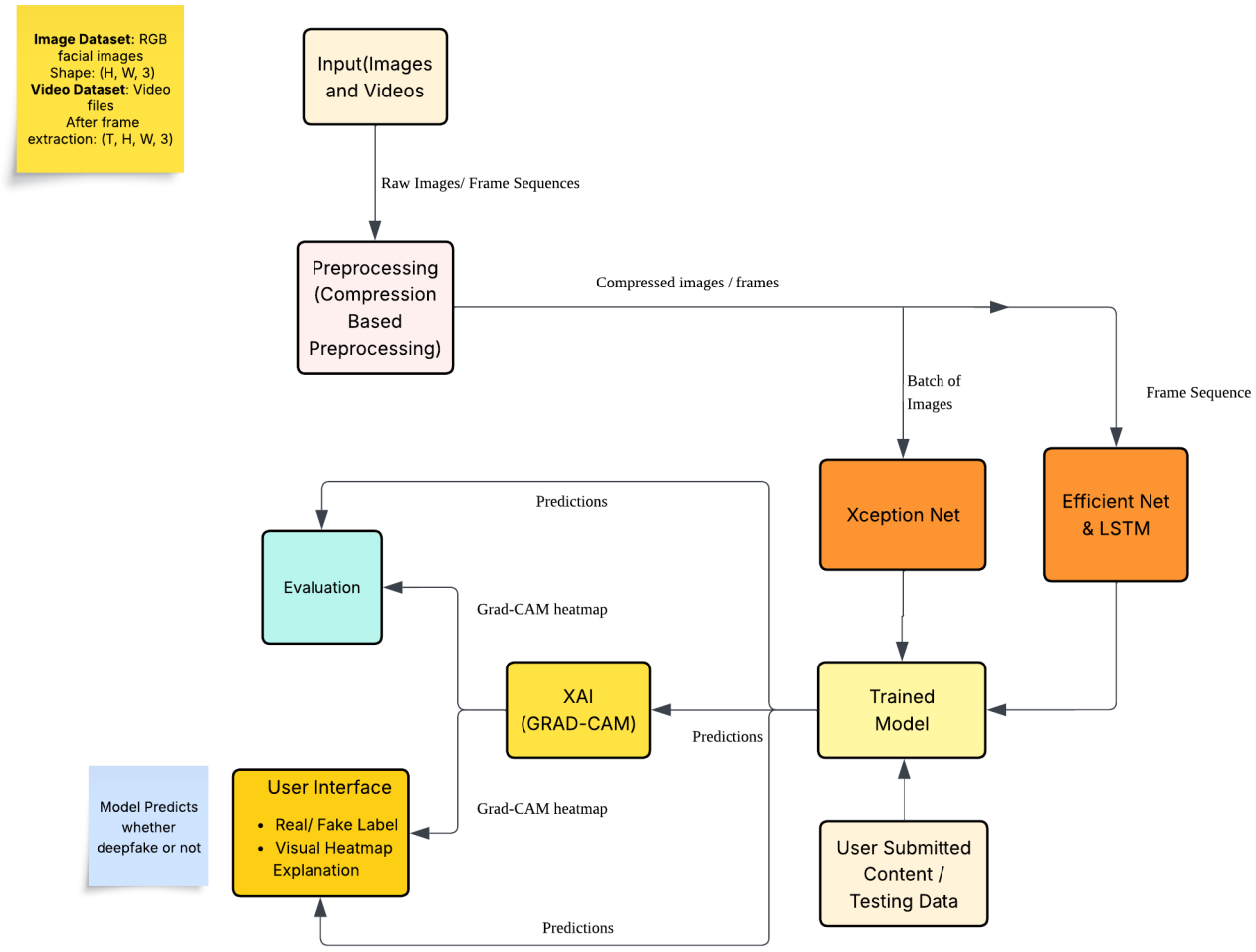


Fig. 3. CADET architecture for compression aware deepfake detection

dimensions to 512 first, following which a ReLU activation function and a dropout layer with probability 0.1 is applied. Following which, the result is given to another fully connected layer that reduces the 512-dimensional feature vector into a singular logit value. Lastly, the logit is passed through the sigmoid function σ to convert it into a prediction value $\hat{y}_{img}$, representing the model's prediction of if the given image is a deepfake or not.

$$\hat{y}_{img} = \sigma\left(\mathbf{W}_2 \cdot \text{Dropout}\left(\text{ReLU}\left(\mathbf{W}_1 z_{img} + \mathbf{b}_1\right)\right) + \mathbf{b}_2\right) \tag{5}$$

H. Detection - Video

The feature extractor chosen for the video model is EfficientNet-B1 because it prioritizes state-of-the-art accuracy in image classification tasks while using fewer parameters than similar models [23]. The B1 model in particular is selected due to its comparable classification performance to Xception [24]. An LSTM was used for temporal feature extraction due to its ability to retain context across long sequences [25]. The input to the video part of CADET is given to the pretrained EfficientNet-B1 backbone which acts as a spatial feature extractor and outputs a 1280-dimensional feature vector. This is then passed to the LSTM which extracts temporal features from across frames using the spatial features. The LSTM is uni-directional and uses 2 layers with a hidden layer size of 512 and a dropout layer with probability of 0.1. We average the feature values from the LSTM across all frames to produce a single 512-dimensional feature vector z_{vid}.

$$z_{vid} = \frac{1}{T}\sum_{t=1}^{T} \text{LSTM}_t\Big(\text{EfficientNet}(X_t)\Big) \in \mathbb{R}^{B_{vid} \times h} \tag{6}$$

with parameters:

- T represents the number of frames
- B_{vid} represents the batch size (8 for video)
- h represents the hidden layer size of the LSTM
- X_t represents the input tensor values at frame t

z_{vid} is then given to a custom classifier head to produce the raw logits. The classification head for the video model

consists of a fully connected layer that converts the 512-dimensional feature vector into a 256-dimensional vector, following which a ReLU activation function and a dropout layer with probability 0.1 are applied. This result is then given to another fully connected layer that produces a singular raw logit. The raw logit is passed into a sigmoid activation function σ that converts the raw logit into a prediction value $\hat{y}_{vid}$, representing the model's prediction of if the given video is a deepfake or not.

$$\hat{y}_{vid} = \sigma\left(W_4 \cdot \text{Dropout}\big(\text{ReLU}(W_3 z_{vid} + b_3)\big) + b_4\right) \quad (7)$$

I. Explainability

The explainability technique used for this work is Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM provides improved computational efficiency over perturbation based methods like LIME or SHAP while not requiring a change in the model architecture like regular CAM. Grad-CAM was used to highlight the regions of the image/frame that are important to the model by computing the gradient of the output with respect to the activation A that occurred at the region for each feature.

$$L^t_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^t_k A^t_k\right) \in \mathbb{R}^{H \times W} \quad (8)$$

where the feature weight α^t_k for feature k at frame t is computed via global average pooling of the gradients:

$$\alpha^t_k = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{\partial y}{\partial A^t_{k,i,j}} \quad (9)$$

V. Results and Discussion

The video detection part of the system was tested using the FaceForensics++ dataset, using videos from five different manipulation types, Deepfakes, Face2Face, FaceShifter, FaceSwap, and NeuralTextures, along with real videos. In total, 1000 real and 5000 fake videos were taken, and from each video, sequences of 16 frames were pulled out for processing. These were then divided into training and test sets. Training was done over 5 epochs using the AdamW optimizer, with a learning rate of 0.0001 and a batch size of 8. Every frame was resized to 224×224 before being passed into the model, and compression based preprocessing was applied.

Looking at Figure 4, the training loss does come down over time while the validation loss reaches its lowest at epoch 2 with increasing validation loss over further epochs. The validation accuracy also seems to stagnate after epoch 2, reaching a maximum at epoch 4 with a value of 84.00%.

TABLE I
METRICS FOR VIDEO DETECTOR

Metric	Percentage
Accuracy	79.00
Precision	79.03
Recall	79.00
F1 Score	78.99

Table I breaks down how the model performed on the test set. It got an accuracy of 79.00%, a precision of 79.03%, a recall of 79.00%, an F1-score of 78.99%, and an AUC-ROC of 0.8703. Despite being trained on unbalanced datatset, the model has clearly learned to distinguish between real and deepfake videos according to the high F1-score. Lastly, the high AUC-ROC indicates the model's overall predictive capacity across every decision boundary is also good.

Figure 6 shows some of the Grad-CAM heatmaps the system produced for a fake sample. In the visualization, the highlighted areas were mainly around the eyes and mouth, which is expected since these are the regions most commonly tampered with during deepfake creation. This is exactly what you would want from an explainability module, because instead of just telling someone a video is fake, the system is actually pointing to where the problem is. Furthermore, the Grad-CAM heatmaps were subject to deletion based faithfulness testing [26] which showed model confidence degradation with increasing proportion of deleted significant pixels as well as a AUC_{del} value of just 0.4579 for the video model.

Taking everything together, the numbers show that detecting all deepfakes under heavy compression is genuinely hard, and the model still has potential for improvement.

The image detection component of the system was evaluated using the 140k Real and Fake Faces dataset. The dataset provides a balanced distribution containing 70,000 authentic facial images from Flickr and 70,000 synthetically generated fake faces produced via StyleGAN. The data was rigorously partitioned into training, testing, and validation splits according to a 70:15:15 ratio. The spatial deepfake detector was built upon a pre-trained XceptionNet core with a modified final classification head for binary outputs with an integrated 10% dropout rate to mitigate overfitting. Training was executed over 10 epochs using the Adam optimizer with a batch size of 32, a fixed learning rate of 0.0003, and binary cross-entropy loss with logits (BCEWithLogitsLoss) as the primary loss function. All images were uniformly resized to 299 × 299 pixels and subjected to the dynamic Compression-Based Preprocessing (CBP) pipeline to ensure high robustness against social media compressions.

TABLE II
METRICS FOR IMAGE DETECTOR

Metric	Percentage
Accuracy	99.76
Precision	99.78
Recall	99.73
F1 Score	99.75

Table II shows the performance metrics of the Xception-based image detector. Even with real-world compression, the model worked very well. It achieved an accuracy of 99.76%, precision of 99.78%, recall of 99.73%, and an F1-score of 99.75% . The image model successfully found tiny grid patterns and blending mistakes left behind by the StyleGAN architecture. This is different from the video model, where

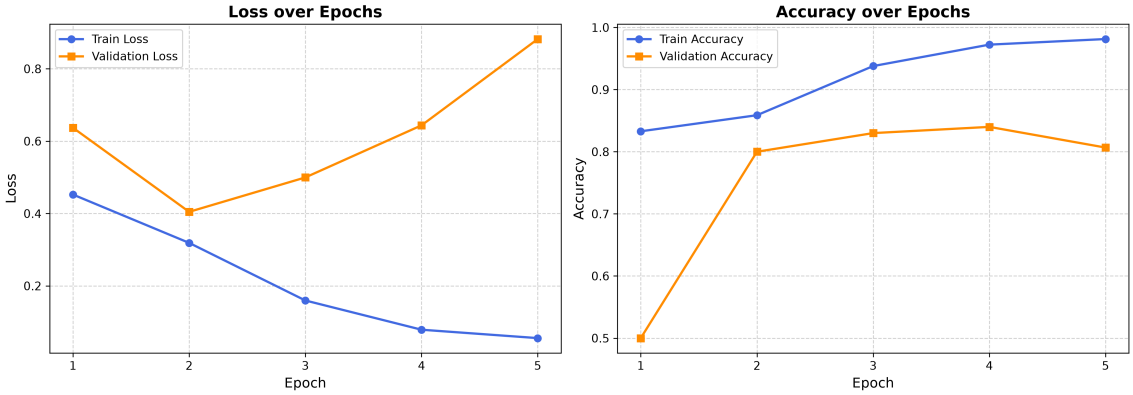


Fig. 4. CADET video model training and validation metrics over epochs.

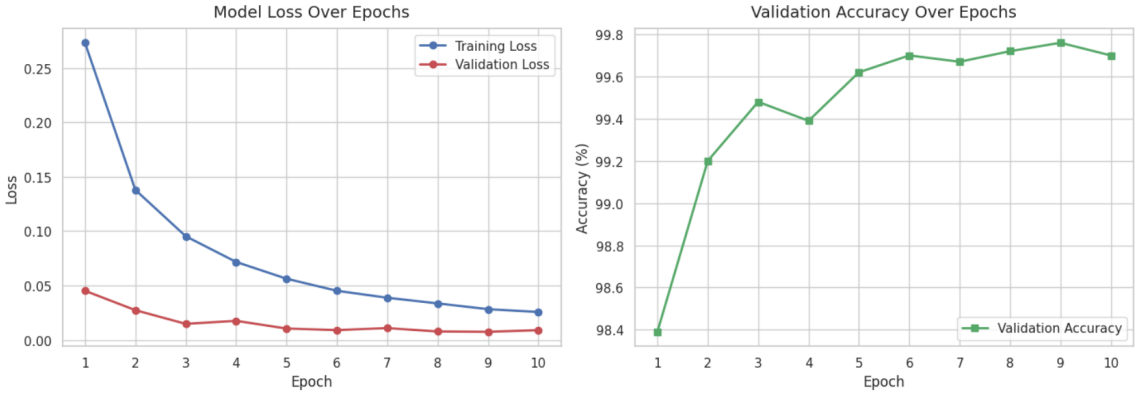


Fig. 5. CADET image model training and validation metrics over epochs.

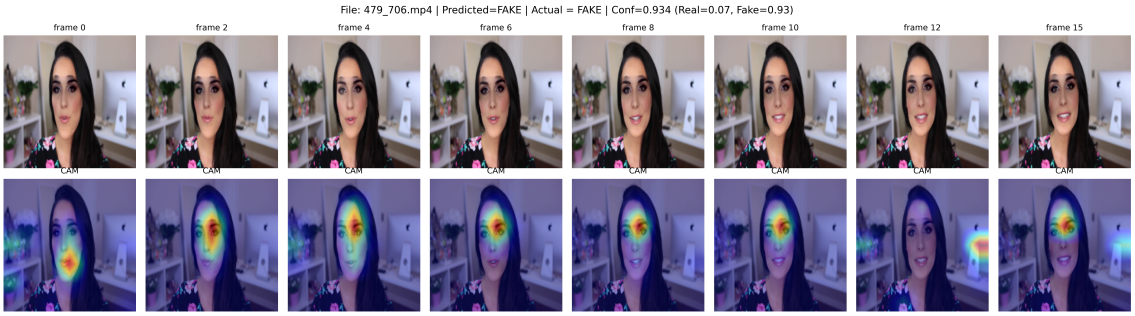


Fig. 6. Sample CADET Grad-CAM heatmap outputs for real and fake video frames.

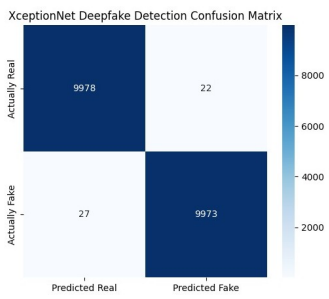


Fig. 7. Confusion matrix of the CADET image model.

heavy compression often removes small details across video frames. Given the exceptionally high accuracy of 99.76% on the image dataset, we closely evaluated the model for potential overfitting. We confirmed that the model is generalized and stable due to two key factors. First, the integration of a 10% dropout rate in the custom classification head successfully prevented the network from over-relying on specific node paths, forcing it to learn more robust features. Second, the 99.76% accuracy was achieved on a completely independent test split that the model never encountered during the training phase.

VI. Conclusion

Most deepfake detectors today give you a prediction score but never explain how they got there, this work was built to change that. The system brings together a compression based preprocessing pipeline to handle the kind of media degradation that happens on social media, XceptionNet for catching manipulated images trained on the 140k Real and Fake Faces dataset, and EfficientNet with LSTM for handling videos evaluated on the FaceForensics++ dataset. On the image side, the model achieved an accuracy of 99.76%, and on the video side it achieved 79.00%. Going beyond a simple real or fake output, the system was designed with the understanding that analysts need more than just a prediction to take decisions. Through Grad-CAM, it generates visual heatmaps that point out exactly which facial regions or video frames gave it away, giving analysts something concrete to look at rather than just a number to trust. Testing also showed that the system holds up well even when the media has gone through heavy compression. In the end, this paper tackled three problems that keep coming up in existing deepfake detection work, which are the inability to handle compression, the lack of systems that work for both images and videos, and the absence of any real explanation behind the predictions.

VII. Future Enhancements

Although the system built in this work is a good step toward making deepfake detection more transparent and reliable, there are still several areas where it can be taken further. One possible avenue is bringing in Adversarial Robustness Training(ART) [14]. Right now, the system handles compressed media well, but it has not been hardened against adversarial attacks. Adding ART into the pipeline would make the system much harder to fool. Another area that needs attention is how well the system holds up across different datasets.

At the moment, the models are trained and tested on specific datasets, and as a result, performance can drop when the model encounters deepfakes made using methods it has never seen before. Going forward, improving the model's ability to generalize across different datasets and generation approaches is something that needs to be worked on.

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